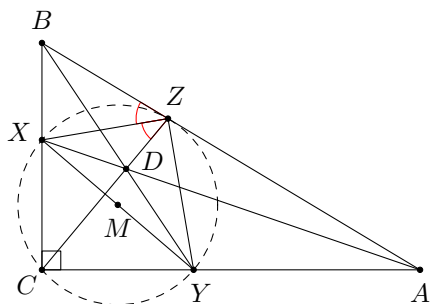


Problem 21 (Thailand MO 2011 Problem 3). In $\triangle ABC$, $\angle C = 90^\circ$. A point D lies inside $\triangle ABC$. The lines AD, BD and CD intersect BC, CA and AB at X, Y and Z respectively. Let M be the midpoint of XY . If XZ bisects $\angle BZC$, prove that $MZ = MC$.



Solution. By Ceva's theorem, we have $\frac{BX}{XC} \times \frac{CY}{YA} \times \frac{AZ}{ZB} = 1$. Note that $\frac{BX}{XC} = \frac{BZ}{ZC}$ by the angle bisector theorem. So this implies $\frac{CY}{YA} = \frac{CZ}{ZA}$.

Again by the angle bisector theorem, this yields $\angle CZY = \angle AZY$. Thus, $\angle XZY = \frac{1}{2} \angle BZA = 90^\circ$. It follows that C, Y, Z, X are concyclic, and M is the centre of $(CYZX)$ since XY is a diameter of this circle. This proves $MC = MZ$. $\square$

Problem 22 (Canada MO 2013 Problem 2). Find all positive integers n for which there exists a permutation $a_1, a_2, \dots, a_n$ of the numbers $1, 2, 3, \dots, n$ such that the remainders of

$$0, a_1, a_1 + a_2, \dots, a_1 + a_2 + \dots + a_n$$

when divided by $n + 1$ are pairwise distinct.

Solution. The answer is any odd positive integer n .

If n is even, then $a_1 + a_2 + \dots + a_n = 1 + 2 + \dots + n = \frac{n(n+1)}{2} \equiv 0 \pmod{n+1}$. Therefore, two of the terms have the same remainder.

If n is odd, consider the permutation $1, n-1, 3, n-3, \dots, n-2, 2, n$ of $1, 2, \dots, n$. Then it is easy to see that

$$0, a_1, a_1 + a_2, \dots, a_1 + a_2 + \dots + a_n \equiv 0, 1, -1, 2, -2, \dots, \frac{n-1}{2}, -\frac{n-1}{2}, \frac{n+1}{2} \pmod{n+1}.$$

The remainders are pairwise distinct. $\square$

Problem 23 (Kürschák Competition 2014 Problem 3). Let $\mathcal{P}$ be a convex polygon on the plane. Prove that for any point on the plane, we can reflect the point in the sides of $\mathcal{P}$ for a finite number of times such that the image lies on or inside $\mathcal{P}$.

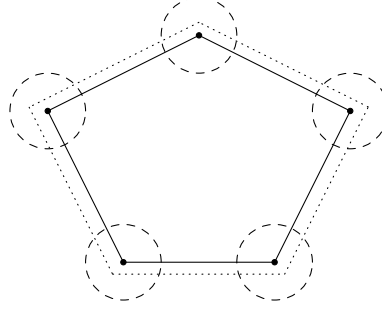
Solution. We say that a region is *good* if every point lying on or inside this region satisfies the desired property. We need to prove the whole plane is good.

Claim 1: There exists $r > 0$ such that every disc with centre being a vertex of $\mathcal{P}$ and radius r is good.

Let V be a vertex of $\mathcal{P}$ and let θ be the interior angle of $\mathcal{P}$ at V . Firstly, it is clear that there exists a sector with centre V , positive radius and angle θ that lies in $\mathcal{P}$. This sector must be good. Reflecting this sector in one of the sides with endpoint V , we obtain a good sector with angle 2θ . Next, note that the composition of the reflection in a side with endpoint V and the reflection in another side with endpoint V is a rotation with centre V and angle 2θ . Thus, by rotating the sector about V by angle 2θ for a finite number of times, we eventually obtain a disc with centre V and positive radius which is good. Since there is a finite number of vertices, we can take the minimum such radius to be r so that claim 1 holds.

Claim 2: There exists $\varepsilon > 0$ such that the following holds. If we translate each sideline of $\mathcal{P}$ away from $\mathcal{P}$ for a distance of ε , then the region bounded by the images (called the ε -region) is good.

Firstly, we choose $r > 0$ by claim 1 and draw a good disc using each vertex of $\mathcal{P}$ as centre and radius r . Consider a sufficiently small $\varepsilon > 0$ and consider the ε -region. We reflect $\mathcal{P}$ in each of its sides. Note that all these images are good. The union of these images covers the ε -region, possibly except some regions which are sufficiently close to one of the vertices of $\mathcal{P}$ (when some interior angles are less than 120°). But these exceptional regions are contained in the good discs. Therefore, the ε -region is also good.

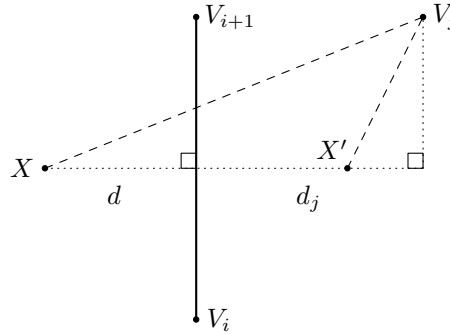


Claim 3: For any point X outside $\mathcal{P}$, let $c = XV_1^2 + XV_2^2 + \dots + XV_n^2$ where $V_1, V_2, \dots, V_n$ are the vertices of $\mathcal{P}$. There exists a sideline V_iV_{i+1} (with $n+1=1$) such that X and $\mathcal{P}$ lie on different sides of V_iV_{i+1} . Let d be the distance from X to V_iV_{i+1} , and let d_j be the distance from V_j to V_iV_{i+1} . Then c is decreased by $4d(d_1 + d_2 + \dots + d_n)$ if we reflect X in the side V_iV_{i+1} .

Let X' be the image of X . By Pythagoras' theorem, we have

$$XV_j^2 - X'V_j^2 = (d_j + d)^2 - (d_j - d)^2 = 4d_jd$$

for each j . The result follows readily.



Now, suppose a point X does not lie in any good region. Using claim 3 repeatedly, we can reflect the point indefinitely such that $c = XV_1^2 + XV_2^2 + \dots + XV_n^2$ is decreased in each reflection (the process can be repeated since we have assumed all images cannot lie on or inside $\mathcal{P}$). As $c \geq 0$, it must converge to some limit. This implies the quantity $4d(d_1 + d_2 + \dots + d_n)$ defined in claim 3 converges to 0. As $d_1, d_2, \dots, d_n$ are constants, this implies d converges to 0. Therefore, after some steps, we must have $d < \varepsilon$, where ε is chosen by claim 2. This shows the image of X lies inside the ε -region, which is good by claim 2. This is a contradiction. Therefore, the whole plane is good. $\square$

Problem 24 (Kürschák Competition 2016 Problem 3). Let $P(x)$ be a polynomial with real coefficients such that $P(P(x))$ is the square of a polynomial with real coefficients. Is it true that $P(x)$ is the square of a polynomial with real coefficients?

Solution. Yes. Let $\deg P = d$. The result is obvious if $d = 0$. Suppose $d \geq 1$. We have $\deg(P(P(x))) = d^2$. Since $P(P(x))$ is a square, d^2 is even. Let $d = 2k$. Let a be the leading coefficient of P . Then the leading coefficient of $P(P(x))$ is a^{2k+1} . As $P(P(x))$ is a square, a must be positive.

Claim: There exists a polynomial $Q(x)$ of degree k with real coefficients such that $\deg(P(x) - Q(x)^2) \leq k - 1$.

Let $P(x) = \sum_{i=0}^{2k} a_i x^i$ and consider $Q(x) = \sum_{i=0}^k b_i x^i$. Then we have

$$P(x) - Q(x)^2 = \sum_{i=0}^{2k} \left(a_i - \sum_{j+j'=i} b_j b_{j'} \right) x^i.$$

It suffices to choose $b_k, b_{k-1}, \dots, b_0$ one by one so that $a_i - \sum_{j+j'=i} b_j b_{j'} = 0$ for $i = 2k, 2k-1, \dots, k$. Indeed, we have $b_k = \sqrt{a_{2k}} > 0$ (which is real since $a_{2k} > 0$), and

$$b_{i-k} = \frac{1}{2b_k} \left(a_i - \sum_{\substack{j+j'=i \\ j, j' \neq k}} b_j b_{j'} \right)$$

for $i = 2k-1, 2k-2, \dots, k$.

Choose $Q(x)$ as in the claim. Let $P(x) = Q(x)^2 + R(x)$ where $\deg R \leq k-1$, and let $P(P(x)) = S(x)^2$. Then

$$[S(x) - Q(P(x))][S(x) + Q(P(x))] = S(x)^2 - Q(P(x))^2 = P(P(x)) - Q(P(x))^2 = R(P(x)). \quad (1)$$

As $\deg S = \deg(Q(P(x))) = 2k^2$, at least one of $S(x) - Q(P(x))$ and $S(x) + Q(P(x))$ has degree $2k^2$. However, we see that $\deg(R(P(x))) \leq 2k(k-1) < 2k^2$. Therefore, the only possibility is that both sides of (1) are 0. In particular, $R(P(x)) = 0$, and hence $R(x) = 0$ (as P is non-constant). This implies $P(x) = Q(x)^2$ as desired. $\square$